# Paulson Alex Fernandes

*Cybersecurity & Digital Forensics Student | SOC & Threat Detection | Python + ML*

+918459150386 paulsonfernandes.dev@gmail.com LinkedIn GitHub Website Pune, India

## WORK EXPERIENCES

**Cybersecurity Intern** | Internship — Aug 2025 - Aug 2025

*Btech Walleh* — *Remote*

- Investigated **Conti ransomware** in Splunk, identifying persistence mechanisms, credential-theft indicators, and exploited CVEs.
- Analyzed **Lumma Stealer** activity using Wireshark and Zeek to identify malicious domains, C2 traffic, and data-exfiltration indicators.
- Conducted controlled Windows penetration tests using **Metasploit** and **EternalBlue**, including privilege escalation and password-hash cracking.
- Simulated ARP poisoning and configured Windows Firewall rules to demonstrate network risks and strengthen endpoint defenses.

***Technologies / Skills Used :*** *Splunk, Wireshark, Zeek, Metasploit*

## PROJECTS

**AI Malware Detection System** | Github

*Developer*

- Built a static-analysis pipeline to classify Windows executables using 50+ structural, entropy, and import/export features.
- Developed model comparison, explainability, scan-history tracking, and downloadable PDF security reports.

***Technologies / Tools Used :*** *Python, Flask, Random Forest, XGBoost, SQLite, ReportLab*

**HoneyWatch – SSH Threat Intelligence Honeypot** | Github

*Developer*

- Designed an SSH honeypot with stateful fake-shell emulation and canary credentials to capture attacker behavior and malware samples.
- Implemented GeoIP enrichment, attack-pattern detection, and real-time WebSocket monitoring of attacker origins.

***Technologies / Tools Used :*** *Python, Paramiko, asyncio, SQLite, WebSockets, GeoIP*

**VaultIX – Secure Password Manager** | Github

*Developer*

- Developed a browser-based password vault with client-side AES-256-GCM encryption and PBKDF2 key derivation.
- Implemented cryptographically random password generation, entropy-based strength scoring, auto-lock, and secure import/export.

***Technologies / Tools Used :*** *JavaScript, Web Crypto API, AES-256-GCM, PBKDF2*

**FORGE-X | Digital Forensics Evidence Management System** | Github

*Developer*

- Developed a web-based digital forensics management system using **Flask and MySQL** to manage investigation cases, digital evidence, examinations, and forensic reports.
- Implemented **SHA-256 evidence integrity verification**, chain-of-custody tracking, role-based access control, and audit logging to support evidence traceability.
- Designed a normalized relational database with constraints, views, stored procedures, triggers, and transaction management to enforce data integrity and business rules.

***Technologies / Tools Used :*** *Python, Flask, MySQL, HTML, CSS, JavaScript*

## SKILLS

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|---|---|
| **Databases :** | MySQL, SQLite |
| **Frameworks & Libraries :** | Scikit-learn, XGBoost, Random Forest |
| **Languages :** | English, Hindi, Konkani |
| **Programming Languages :** | Python, C++, JavaScript, C |
| **Soft Skills :** | Communication, Teamwork, Leadership, Problem Solving, Decision Making |
| **Tools & Platforms :** | Splunk, Wireshark, Zeek, Metasploit, Nmap, Microsoft Sentinel, Microsoft Defender, Ettercap, John the Ripper, Linux |

## EDUCATIONS

**MIT World Peace University** | Pune — 2025 - 2029

*B.Tech, CSE (Cybersecurity and Forensics)*

## CERTIFICATIONS

- **HackerRank**: Software Engineer Intern (Sep 2026)
- **Palo Alto Networks Academy**: Cybersecurity Fundamentals (Nov 2025)
- **Palo Alto Networks Academy**: Network Security Fundamentals (Apr 2026)
- **Launched Global**: Cyber Security Mentorship Program (Certificate of Course Completion)

## LEADERSHIP & LANGUAGES

**Liaison Officer** — Sep 2025

*Smart India Hackathon 2025 (Internal)*

- Coordinated teams, mentors, and organizers.

*Languages*

- English (**Full Professional**)
- Hindi (**Native**)
- Konkani (**Native**)